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On derived domains in sentence phonology

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I Introduction

This paper deals with two distinct but inextricably connected sets of questions in the area of sentence phonology. The first concerns the organisation of sentence phonology and the nature of the phonological representation(s) of the sentence, and the second the relation between syntactic structure and phonological representation.

With respect to the first, the most fundamental question concerns whether there are distinct levels of representation mediating between surface syntactic structure, the input to the sentence phonology, and phonetic representation, its output:

(I) surface syntactic structure



phonetic representation

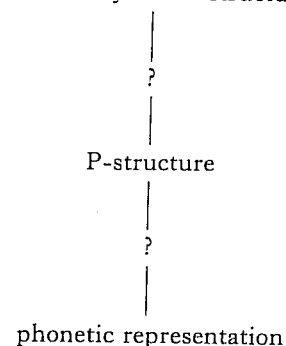
Other important related questions are the following. If intermediary levels

do exist, do the representations at all these levels have the same, or different, formal properties? Do the rules applying in the different subcomponents corresponding to these levels display distinct syndromes of properties? Or is sentence phonology homogeneous?

The first part of this paper contributes to this area of inquiry in arguing that there is a level in sentence phonology at which the representation is organised hierarchically into what have been called prosodic constituents (of larger-than-word size) and that there are phonological rules (of various types) which have their characteristic domains defined in terms of this prosodic structure. This is a point that has been made in previous work (Selkirk 1978, 1980a, b, 1984; Nespor & Vogel 1982, 1986; Hayes 1984; Booij 1983; Chen 1985); here I am bringing new evidence to bear on the issue. I will argue for a new analysis of the distribution of vowel length in Chi Mwi:ni sentences which entails that Chi Mwi:ni sentences be organised into a prosodic constituent structure. Chi Mwi:ni is a language closely related to Swahili that is spoken in the town of Brava in Somalia. The data used here is drawn from descriptions by Goodman (1967), Kisseberth & Abasheikh (1974) and Kenstowicz & Kisseberth (1977). It will be argued that the pattern of distribution of vowel length within the phrasal domain provides evidence for the assignment of a pattern of stress on that domain (one that is extremely familiar from studies of word stress: antepenultimate stress unless penult is heavy, penultimate stress otherwise) and for a shortening rule which simply affects long vowels that are stressless. This stress-based analysis of the distribution of vowel length in the Chi Mwi:ni sentence has interesting implications for the theory of phonology. Because the 'word stress module' (involving a set of parameters such as laid out in work by Halle & Vergnaud 1978 and Hayes 1980 in a metrical tree framework and Prince 1983 and Selkirk 1984a in a metrical grid framework) is available in the sentence phonology in Chi Mwi:ni, clearly there is no principled connection that can be drawn in the theory of grammar between this type of patterning and the component in the grammar where it is defined. Moreover, and more important for our concerns here, while the domain of application of 'word stress patterns' is typically a syntactic constituent, one that is word-sized or smaller, in Chi Mwi:ni the domain is 'phrasal' (in a sense to be defined), and even more interesting, it is demonstrably not a syntactic constituent. Since stress rules require the representation of end-of-domain (their 'point of demarcation', so to speak), it is argued that these facts force us to assume that derived (phonological) domains are given a representation in the phonological representation of the sentence, of the sort envisaged in the theory of prosodic structure.

From the organisational standpoint, then, the argument is that there exists a level, which I will call P-structure, mediating between syntactic structure and phonetic representation. This P-structure is constituted of, among other things, prosodic structure, and certain phonological rules apply in virtue of that structure:

(2) surface syntactic structure



The second set of questions concerns the relation between syntactic structure and phonological representation. Does syntax impinge on sentence phonology throughout the derivation, or is its effect restricted to only certain levels? Is a general theory possible of what aspects of syntactic structure are available to rules of sentence phonology, or might this vary according to what type of rule or what subcomponents are involved?

In the second part of this paper I argue for a theory of just how prosodic structure is constituted on the basis of syntactic structure. It turns out that the properties of syntactic structure which find themselves reflected in prosodic structure are extraordinarily limited. As we will see, in defining this mapping reference is required only to the right or left END of syntactic constituents of designated types in the X-bar hierarchy (cf. Jackendoff 1977; Stowell 1981; Emonds 1985). The original insight that the mapping of syntactic structure into prosodic structure is END-BASED is due to Chen (1985), and is developed in the course of this paper. Such a result is clearly of interest in the study of the on-line processing of sentences: ascertaining the limit of a prosodic constituent of a particular sort will mean (depending on the language involved) finding the right or left limit of a syntactic constituent of a particular level in the X-bar hierarchy, surely a useful bit of information in the assigning of a syntactic constituent structure to a sentence.

Prosodic structure, then, in its very constitution, encodes a certain amount of information about the syntax. A hypothesis I have been entertaining for some time is that the level at which prosodic structure forms part of the representation is one where there is no syntactic structure (any more). In other words, once the translation of syntax into prosodic structure is made, the syntax itself falls out of the picture. On this view, the level of P-structure is something of a syntax/phonology watershed. Rules in the subcomponent(s) before it may access the syntax (in ways to be defined); rules in the subcomponent(s) after it may not.

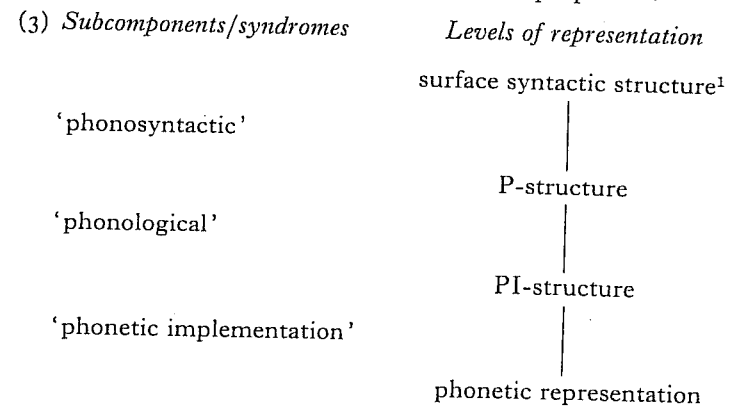
If we admit the existence of rules in the pre-P-structure part of the

grammar which may be directly syntax-sensitive and rules in the post-P-structure part which may be indirectly syntax-sensitive, via their sensitivity to syntax-dependent prosodic structure, then the question will arise for any seemingly syntax-sensitive rule whether it belongs in the former or the latter category. Presumably a narrow enough characterisation of the possible influence of syntax in the pre-P-structure 'phonosyntactic' mode(s) and a narrow enough characterisation of the mapping between syntactic and prosodic structure and of the way in which post-P-structure rules are sensitive to prosodic structure will permit one to answer this question. (Other properties of the rules besides those involving domain might permit ascribing them to one or another subcomponent. For example, perhaps the pre-P-structure rules may retain certain features of the lexical phonology, while the post-P-structure rules do not. What is needed is a full fleshed-out theory of the different syndromes.) In this paper I will be arguing that certain rules of sentence phonology which have been understood to be directly syntax-sensitive should instead be classed as P-structure-sensitive rules. These include rules that Kaisse (1985) and others have argued are subject to a general condition which says, roughly speaking, that among two words in sequence involved in the application of a rule, one must c-command the other.

By including a considerable number of the sandhi rules which have been construed to be syntax-sensitive in the class of P-structure rules, the set of rules presumed to belong to the pre-P-structure 'phonosyntactic' component becomes rather small. It might still include Mende phrasal mutation (Conteh *et al.* 1986), Kimatuumbi vowel shortening (Odden 1982), and perhaps even rules like English *wanna* contraction (see, for example, the discussion in Selkirk 1984: ch. 7), which appear to be sensitive to a richer array of syntactic conditions than would be expressible via the translation of syntax into P-structure. A full theory of this syntactic conditioning is yet to be articulated, however, as is a theory of the substantive phonological properties displayed by these rules.

Before proceeding to the business at hand, I must point out one further articulation of the model of sentence phonology that seems to be in order. Recent studies in phonetics and phonology (Pierrehumbert 1980; Liberman & Pierrehumbert 1984; Pierrehumbert & Beckman 1986; Fourakis 1980; Keating 1985) have led to the conclusion that phonetic rules play a larger role in language-particular descriptions or grammars than was assumed to be the case in the standard theory of generative phonology (Chomsky & Halle 1968), where phonetic rules were presumed to be more or less universal in character. (I am using the term 'phonetic rule' to designate those assigning quantitative values to aspects of the phonological representation.) For example, Pierrehumbert argues that the *F₀* contour is the phonetic representation of intonation in English, derived through the operation of phonetic implementation rules (some language-particular, some not) on the phonological representation of intonation. This result leads us to modify our notion of the phonetic representation

which is the output of the rules of grammar: it will have quantitative values on a variety of dimensions, specified by rules of the language in combination with general principles. If this is phonetic representation, though, and if grammar contains rules of phonetic implementation, then our model of sentence phonology requires elaboration. Can we assume a division into rules of phonetic implementation and rules of the phonology *per se*? Can we assume a level of representation that defines the watershed between these two? Answering in the affirmative, the full model would include at least these four distinct levels of representation and potentially three distinct subcomponents and syndromes of rule-properties:



In Selkirk (1984a, b) I argued that the properties of certain rules of external sandhi (e.g. Spanish nasal assimilation) were best explained if they were understood to be rules of phonetic implementation. Their tempo-sensitivity and variability in domain were shown to follow from assuming, among other things, that these rules have a sensitivity to temporal adjacency (in what I've called 'virtual time structure') in the phonetic implementation of the utterance. For these rules 'time' provides a characterisation of their 'domain'. (The output of such rules may also be gradient.) While I still believe it to be true that some sandhi rules of sentence phonology (perhaps even the majority) belong to the phonetic implementation component, I also now believe that there is a properly phonological mode of application, what I've called in the diagram the 'phonological' or the phonology *per se*. The current paper centres around rules of this latter sort. They are not reported to be tempo-sensitive, or variable in domain, or gradient in effect. What I am arguing here is that these 'properly phonological' rules of the sentence phonology have their domains defined in terms of prosodic structure.

Finally, I should say something about how the metrical grid (Liberman 1975; Liberman & Prince 1977; Prince 1983; Selkirk 1984a, b) fits into this picture. In Selkirk (1984a, b) it was argued that the metrical grid gives representation not only to prominence relations or rhythmic patterning

[illegible]

Holding on to a central role for sdb's in the phonology *per se*, one might argue that the prosodic constituency of P-structure is induced from a phonological representation containing sdb's, and that it is silent demibeat insertion, and not the prosodic structure, which is defined directly with respect to the syntax. Alternatively, one might argue that it is the prosodic structure which is basic in P-structure, and defined directly with respect to the syntax. It is this latter tack that I am pursuing here, and I'll give some justification for it below in §3. Notice that giving prosodic structure this central place in P-structure does not commit us to banning sdb's any direct relation to the syntax. Yet it does permit us to entertain the possibility that sdb's, and other aspects of the metrical grid as well, are defined not with respect to syntactic domains but with respect to prosodic structure instead. Indeed, this sort of dependence of metrical grid construction upon prosodic structure is required in the analysis of Chi Mwi:ni stress to be presented here. It is tempting to pursue this tack, hypothesising that all metrical grid construction is performed on prosodic structure domains, and that prosodic structure determines the abstract representation of (dis)juncture in time represented in Selkirk (1984a) as silent demibeats as well. The implications of seeing these two aspects of the hierarchical representation of sentence phonology to be in general related in this way now need investigation, a task unfortunately beyond the scope of this paper.

The following paragraph from Goodman, the first to describe the intricacies of Chi Mwi:ni vowel length, lays out certain basic elements of the analysis:

Two sources of potential length are illustrated here: underlying length, as in *ma:limu*, and lengthening before certain suffixes, as in *mtana:ni*. Two other sources of potential vowel length are a rule that lengthens vowels before nasal + consonant clusters, shown in (5c), and a rule that gives rise to a long vowel as a result of 'contracting' two short vowels brought together through prefixation, shown in (5d):

- (5) a. ma:limu 'teacher' malimuwe 'his teacher'
 mta:na 'room' mta:nawe 'his room'
- b. mtana:ni 'in (the) room' (< mta:na + ni loc.)
 soma:ni 'don't read' (pl.) (< so:ma + ni pl. imper.)
- c. nu:mba 'house' wa:ntu 'people'
- d. ki:wa (< ku+iwa) 'to know' se:leze (< si+eleze) 'don't explain'

Kisseberth & Abasheikh show that all these potentially long vowels are submitted to shortening in the environments described above. (6) gives examples of shortening in pre-antepenultimate position or further back. As the examples reveal, the notions penultimate, antepenultimate and pre-antepenultimate must be defined with respect to some larger-than-word domain:

- (6) a. ma:limu 'teacher' malimu wa sitta 'the sixth teacher'
 ma:yi 'water' mayi malada 'fresh water'
 b. soma:ni 'don't read' somani chuwo ichi 'don't read this book'
 c. chi:ntu 'thing' chintu shpiya 'something new'
 d. ku:la (< ku + ula) kula shta:wa 'to buy a clay pot'

When, through phrasal combination, the penultimite or antepenultimate vowels of the left-hand column find themselves in pre-antepenultimate position or further back, they shorten.

Kisseberth & Abasheikh, who underscore the point that the domain of shortening is phrasal, posit two distinct rules to account for the shortening discussed so far: Pre-antepenult Shortening and Pre-length Shortening. Hayes formulates these as (7a) and (7b):

- (7) a. $V \rightarrow V / - C_0 VC_0 VC_0 V$ Domain: P-phrase
 b. $V \rightarrow V / - \dots \left. \begin{array}{c} VC \\ V: \end{array} \right\}]_{\text{SYL}}$ Domain: P-phrase
 c. $V \rightarrow V / -]_{\text{P-phrase}}$

Note that the domain for these rules is presumed by Hayes to be the prosodic constituent Phonological Phrase. The notion pre-antepenultimate or further to the left is captured in (7a) by the requirement that within the P-phrase there be at least three vowels following the shortened vowel.

There is yet a further source of vowel length, and a further context of shortening, visible only in phrasal contexts. A word-final vowel is always potentially long, and, like the other potentially long vowels, will be realised as such if in the relevant phrasal domain it is either penult or antepenult and not followed by a heavy syllable. The words of the left column in (8) show this final lengthening in the right column:

- (8) xso:ma 'to read' xsoma:chuwo 'to read a book'
 ka:sha 'to light' kasha:nt^ha 'to light a candle'
 hujo 'the one who eats' hujo:mbele 'the one who eats first'
 ni 'is' ni:luti 'it is a stick'

I assume, with my predecessors, that there is a word-final lengthening rule (part of the word phonology) which brings all word-final vowels into the sentence phonology as long:

- (9) *Word-final Lengthening (WFL)*

$$V \rightarrow V: / -]_{\text{Word}}$$

Within the relevant phrasal domain, they are subject to shortening by (7a) and (7b). The examples in (10) show the effects of this shortening:

- (10) xsoma chuwo 'to read his book' (vs. xsoma:chuwo)
 malimu wa sitta (vs. malimu wa:ne 'the fourth teacher')

The absence in phrase-final position of word-final vowel length, moreover, is taken to show the existence of yet another rule of shortening, which Hayes formulates as (7c).²

In short, previous analysts have posited three distinct rules of shortening in the phrasal domain in order to account for the disappearance of 'potential' vowel length on the sentence.

It is significant that the locations for shortening of long vowels in the phrasal domain are just those where a long vowel would never receive (main) stress if Chi Mwi:ni possessed the run-of-the-mill sort of stress system that we see in Latin, for example, applying on the phrasal domain. In a Latin-like word stress system, the final syllable is never stressed, except if the word is monosyllabic. The penult is stressed if it is a heavy syllable (CVV or CVC). Otherwise the antepenult is stressed. Stress never appears in pre-antepenultimate position or further back. These possibilities for stress are illustrated in (11):

(11) *Latin Stress*

$$\begin{array}{ccc} [\dots \acute{\sigma} \sigma \sigma]_{\text{Word}} & \text{vs.} & [\dots \acute{\sigma} \sigma]_{\text{Word}} \\ | & & | \\ \text{CV} & & \left\{ \begin{array}{c} \text{CVC} \\ \text{CV:} \end{array} \right\} \end{array}$$

It should be impossible to avoid the conclusion that Chi Mwi:ni has this same sort of stress system, with a phrasal domain, and a rule like (12) which simply shortens stressless vowels (this latter sort of rule is as commonplace as the Latin-type stress rule):

(12) *Stressless Shortening (SS)*

$$\left[\begin{array}{c} V: \\ - \text{'(main) stress'} \end{array} \right] \rightarrow V$$

Let us review the considerations which force this conclusion.

First of all, this analysis obtains just the right results concerning the distribution of long vowels in Chi Mwi:ni sentences. A phrase-final vowel will not receive stress (it is extrametrical), and so will be subject to shortening. (Here our analysis does the work of (7c).) A vowel in pre-antepenultimate position or further back will never receive stress, and so will shorten. (Here the system is doing the work of (7a).) A long vowel in antepenultimate position will fail to receive stress if followed by a heavy syllable (since the latter receives stress), and so will shorten. (Here the analysis does the work of (7b).) The workings of the analysis are illustrated in the derivations of (13) and (14). The examples of (13) contain underlying long vowels. Each word undergoes WFL. Then, in the sentence phonology, stress is assigned on the appropriate phrasal domain, after which all but the stressed vowels are shortened by SS:

- (13) a. ma:limu b. ma:yi malada
 ma:limu: WFL ma:yi: malada:
 má:limu: Stress ma:yi: málada:
 má:limu SS mayi málada

In (14) the focus is on the retention of the word-final lengthening of the particle *wa*:

- (14) a. ma:limu wa {sitta} b. ma:limu wa saba
 {pi:li}
 ma:limu: wa: {sitta:} WFL ma:limu: wa: saba:
 {pi:li:}
 ma:limu: wa: {sitta:} Stress ma:limu: wá: saba:
 {pí:li:}
 malimu wa {sitta} SS malimu wá: saba
 {pí:li}

When the antepenultimate *wa* precedes a heavy syllable, as in the left column, it will not retain its length, due to SS. But the stressing of antepenult *wa* before the light penult in the right column guarantees the realisation of that length.

A second point is that this stress-based analysis is the only one to explain the precise array of shortening contexts to be found in Chi Mwi:ni. With the earlier analysis, it is an accident that there should be three shortening rules with just these environments. It could equally well have been the case that shortening also took place in penult position, or in penult position only if the antepenult syllable were heavy, etc. The three rules given in (7) in no way constitute a *system*; rather they appear as unrelated phenomena accidentally co-occurring. Assuming there is a system to Chi Mwi:ni shortening, we must reject (7). The stress-based analysis, in contrast, has to recommend it the fact that it does give a systematic account of these shortenings.

Finally, the stress-based analysis should in fact be forced by current phonological theory. In current theorising, no rules, with the possible exception of stress rules, are thought to be given the power to 'count' (Halle & Vergnaud 1978; Hayes 1980; Prince 1983). If stress rules do indeed 'count', then they only count up to two.³ Stress rules may assign a binary alternating pattern. Antepenultimate stress (three from the end of some domain) may be assigned, the argument goes (Hayes 1980, 1982), not because stress rules can count to three, but rather because a domain-final syllable may be extrametrical, and hence ignored, in which case assigning stress two syllables from the (pseudo-)end gives antepenultimate stress. It seems an extremely worthwhile project to pursue such restrictive hypotheses about the nature of the counting that can be performed by phonological rules and about the classes of phonological rules that may be doing that counting, particularly since evidence has not appeared that casts

the theory seriously in doubt. Given current theory, then, the notion pre-antepenultimate should not be a notion that is definable, whether for stress rules or for rules of other types. Consequently the Pre-antepenult Shortening rule of Kisseberth & Abasheikh, formulated as (7a) by Hayes, should not be a possible rule. The theory should provide a single alternative analysis – the stress-based analysis presented here.

Now what of the fact that there is no report in Goodman or Kisseberth & Abasheikh⁴ of 'stress patterns' such as are here being presumed to be in place in Chi Mwi:ni? Perhaps on further investigation 'stress' will prove to be present, with the typical duration, intensity or pitch correlates. But even if no such pattern in the phonetic realisation is discernable, I think it is fair to conclude that Chi Mwi:ni has a stress system of the sort alluded to. What the theory defines at the phonological level are possible patterns of distribution of a particular property, usually called *STRESS*. We could call it *P*. The theory does so through a parameterised system of rules and principles which I have referred to as the 'word stress module'. That module, be it tree-based or grid-based, relies on the notions of *END OF DOMAIN*, *EXTRAMETRICALITY*, *SYLLABLE QUANTITY*, *BINARY ALTERNATION*, etc. That system constitutes a theory of patterns of a particular sort. But by assuming the operation of this *P*-defining module in Chi Mwi:ni, we are not committed to a particular phonetic realisation of *P*. Our stress-based analysis cannot be disproved by showing a lack of patterning in duration, intensity, or whatever. Indeed the proof is in the pattern itself (however it may reveal itself), not in any putative universal instantiation of it.

The conclusion that must be reached, given the considerations above, is that the grammar of Chi Mwi:ni contains a word stress-like system operating on a phrasal domain.⁵ I shall not give a formulation to the system of stress rules in Chi Mwi:ni, partly because the particular formulation, in whatever framework (trees or grid), is irrelevant to the point at hand. What is relevant is that, in any theory of stress patterns, a crucial role is played by domain ends. A pattern of feet (i.e. of stressed and stressless syllables) is always set up from the right or left of some specified domain. Moreover, main word stress is always assigned by an *End Rule*, to use Prince's (1983) term, which seeks out either the first or the last stress within that characteristic domain. Stress rules thus need to know about ends of domains. Which is to say, I would argue, that a bracketing like (15), where α must be fixed for the language at issue, forms part of the structural description of these end-sensitive stress rules, and part of the representation as well:⁶

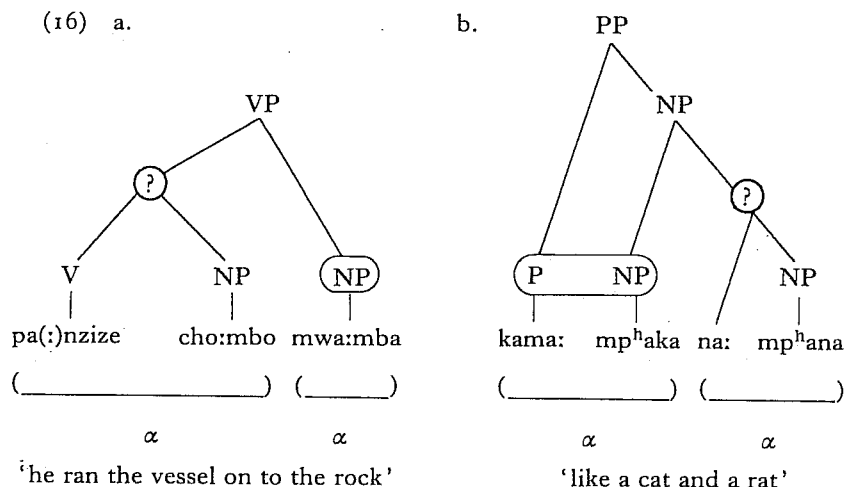
(15) *The representation of domain*

$\alpha[\dots]_{\alpha}$ α = syntactic or phonological category

The Chi Mwi:ni case is interesting not only because the end-sensitive rules of the 'word stress module' are applying on a phrasal, 'non-lexical' domain, but also because the domain does not correspond to a constituent of syntactic structure.⁷ A first important lesson to be drawn from Chi Mwi:ni, then, is that a specification such as (15) of the domain of stress

(and, potentially of other rules) appears in representations and in rules. This point is important as part of the argument I will turn to below against theories like that of Kaisse (1985), for example, which give no place to the representation of derived phonological domains, and therefore no representation of the appropriate ends of domain for Chi Mwi:ni stress. The second lesson is that, since α may designate a category of the phrasal phonology, the crucial notion END must be viewed as quite a general one, not specific to constituents or domains of the syntactic variety.

There are two arguments that can be made that α is not some category of the syntax in Chi Mwi:ni. The first is that there is no one specification for α which gives the appropriate domain for stress. The second, stronger, argument is that the domain designated by α does not always correspond to a constituent of the syntax. The first is illustrated with the example of (16a):



In (16a) the pattern of length shows we must isolate as domains for stress the domains indicated with the bracketing below the tree. Were one not to assume domains as indicated, there would be no explanation for the appearance of the long vowels here, nor for the absence of the underlying (potential) length in the verb. Note that while each of the domains indicated in (16a) does correspond to some possible constituent of the syntax (circled in the tree), there is no consistent theory possible of what the specification for α is, if α in (15) is syntactic. Clearly, α is not a maximal projection, $X^{\max}$. If so, the entire VP would constitute the domain for stress, and there would be no internal domains like those in (16a). Moreover, the evidence is not otherwise consistent with the domain being some putative non-maximal category type. A noun-adjective sequence, for example, is not set off from any following material in the NP.

The point that the value for α is not some syntactic category is

demonstrated by the existence of domains of stress for which there simply is no corresponding syntactic constituent. An example of this is provided in (16b). Again the presence of length in this phrase indicates where the domains must be posited. Perhaps (16b) does not provide the most telling example of a non-constituent domain. One might argue, and this has certainly been done in the literature, that the preposition 'cliticises' to the following NP, the first of the conjoined NPs, and thus does form a syntactic constituent with it. What would decide the question definitively is an example where, for example, a non-function word like a (main) verb occupied the position of P in a structure like (16b). My prediction is that such an example would behave just like the example in (16b), showing that the function word status of the preposition is not what accounts for its being in the same domain with the following NP here. Unfortunately, the data in Kisseberth & Abasheikh is not rich enough to provide evidence on the matter. Yet even in the absence of such direct evidence, I believe that a syntactic cliticisation of the preposition in (16b) can be argued against on general grounds. As I showed in Selkirk (1984a: ch. 7), such cliticisations, motivated on purely phonological grounds, do not in and of themselves explain the phonological behaviour of function words. Other assumptions are necessary, including the extremely general conditions that function words are in no way treated as 'real' words in the grammar (and hence do not receive 'word stress', etc.). But these additional assumptions on their own suffice to explain the phonological behaviour of function words – 'cliticisation' such as that evoked here constitutes a useless extra step, and so, by Occam's razor, is to be avoided.⁸

So we remain with the result that in Chi Mwi:ni the domain of stress corresponds to a sequence in syntactic structure which does not itself correspond to a syntactic constituent. From this we conclude that stress in Chi Mwi:ni has a derived, non-syntactic-constituent domain, a constituent of phonological representation.

3 Deriving prosodic structure

3.1 An end-based theory of derived domains

Kisseberth & Abasheikh make the point that the domain for the distribution of vowel length in Chi Mwi:ni is larger than the word and that it can be defined in rather precise fashion with respect to the syntax, while at the same time not constituting a syntactic constituent itself. Building on Kisseberth & Abasheikh, Hayes, and following him Nespor & Vogel, propose that the derived domain at issue is a constituent of the phonological representation, one they dub the PHONOLOGICAL PHRASE. The proposal is couched within the framework developed in Selkirk (1978, 1980a, b), and in work by Nespor & Vogel, Hayes, and others, according to which the phonological representation consists of a hierarchy of prosodic constituents or categories which may serve as the characteristic domains for phonological rules. This prosodic hierarchy, while not isomorphic to syntactic structure,

(17) *Properties of prosodic structure*

- Taken together, properties (a)–(d) define objects like that in (18) as well-formed in phonological representation:

[illegible]

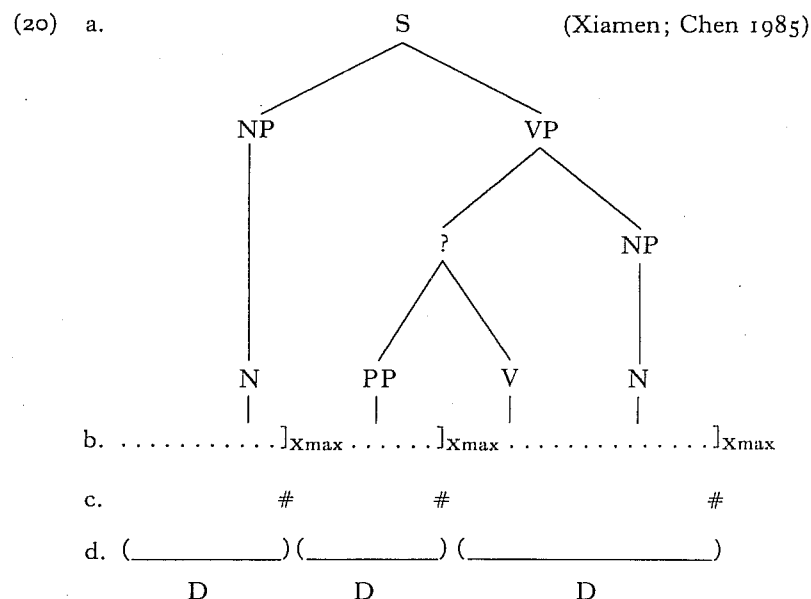
I think it is fair to say that up to now there has been no general theory of the mapping between syntactic structure and prosodic structure to accompany this theory of prosodic structure *qua* representation. And, strictly speaking, no absolutely general theory of the relation is likely to

(19) a. $_{\alpha}[\dots, \text{where } \dots \text{ contains no } _{\alpha}[\dots]$
 b. $\dots]_{\alpha}, \text{ where } \dots \text{ contains no }]_{\alpha}$

The general claim I am making here is that α will be drawn from the set of categories defined in X-bar theory, and that α indicates only a level (or type) in the X-bar hierarchy, which is to say that the syntax-to-prosodic structure mapping is claimed to be cross-categorical in nature. Chen's

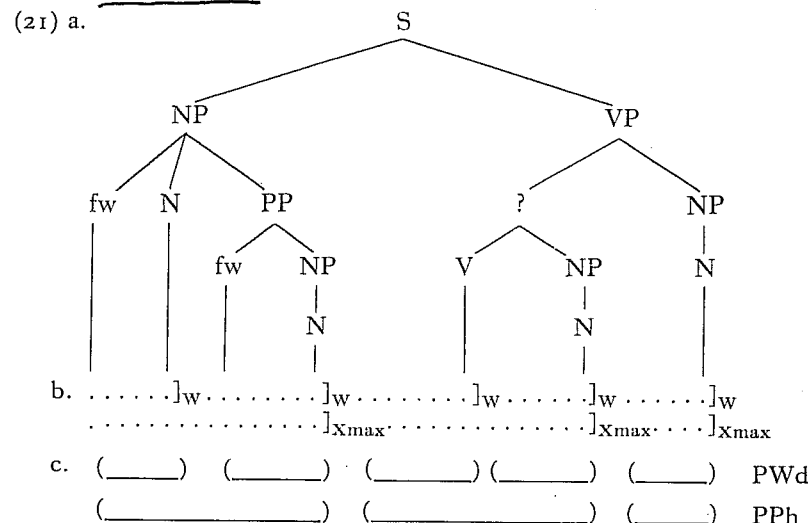
persuasive case for this approach to Xiamen, based on an extremely thorough examination of tone sandhi within a great variety of phrase structure configurations in Xiamen, convinced me of the value of this approach. For Chen, $\alpha = X^{\max}$, and it is the right ends of $X^{\max}$, $]_{X^{\max}}$, that are relevant in Xiamen.

Chen's proposal actually involves the insertion of boundary markers at the appropriate ends, and defines the derived domain as extending between two successive instances of that boundary. Roughly (see below for refinements), the proposal is that at the right end of a maximal projection a tone boundary is inserted. This would give derived domains D for tone sandhi in Xiamen defined as in (20c):



However, since the boundary markers play no other than a domain-delimiting role in the phonology (Chen regards them merely as a mechanical device for defining a prosodic constituent domain as in (20d)), they could just as well be dispensed with, and the prosodic domains defined directly from the appropriate $]_{X^{\max}}$ ends in the syntax. In other words, line (c) can be dispensed with in the mapping of the syntactic structure (20a) into (20d) via the factorisation in (20b) according to the Xiamen end-setting. What is of central importance here is the notion that the syntax-phonology mapping can be defined simply by reference to the ends of syntactic constituents. I would like to suggest in this section that this simple end-based theory of derived domains, appropriately generalised, constitutes both a theory of the relation between syntactic structure and prosodic or phonological structure and a theory of that prosodic structure itself.

First, let us see how the properties of prosodic structure cited above might fall out from the end-based theory. Assume a language with end settings for two derived domains: $]_{\text{Word}}$ and $]_{X^{\max}}$. Applied to a tree like (21), these 'end rules' of the mapping pick out the ends noted in line b:



(Line b is not to be construed as a part of any representation: it only indicates the location of the ends of (21a) which play a role in the mapping into (21c).) Note that function words are not identified as 'real' words, and so do not count in this mapping. They are included in other larger derived domains. Note, moreover, that the domain-formation operates in a bottom-up fashion, operating on the full labelled bracketing, or tree, of the sentence. This will systematically give rise to the possibility that the domains defined are not isomorphic to constituents of the syntax.

A first observation to make about this mapping of syntax into phonology is that it gives rise to two types of domain that are already quite familiar in studies of prosodic structure. In the line derived through the application of the $]_{\text{Word}}$ setting we find domains containing a 'real' word and adjacent function words. This sort of domain is what has been called PROSODIC/PHONOLOGICAL WORD, or CLITIC GROUP,⁹ hence the PWd at the end of that line. In the other line, derived through the $]_{X^{\max}}$ setting, we find the sort of domain that has been dubbed PHONOLOGICAL PHRASE. In view of this, I would like to suggest that a theory of the syntax-phonology mapping based on such end-settings itself provides a theory of possible prosodic constituents (above the foot and below IPh). The hypothesis is that there are only as many prosodic levels as there are possible settings for the label on the end bracket. Here, then, the end-based theory provides a theory of the first property, (17a), of prosodic structure.

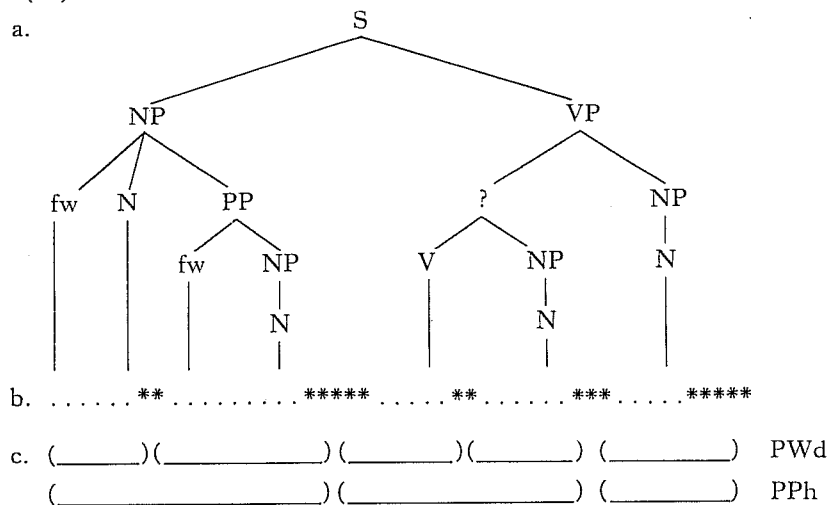
As (21c) shows, the second property, (17b), also falls out from this theory of the mapping. The sentence is exhaustively parsed into domains

of the two types. As for the third property, the hierarchy of domains and their strict organisation into levels is given by the hierarchy of syntactic categories themselves (in the end-settings), and by the simple-minded way the tree is scanned for instances of these ends. Finally, what we have here in (21c) is a well-formed bracketing, called for in (17d).

It is an extremely important result if, on further examination, all these properties of prosodic structure do indeed fall out from an end-based theory of the mapping into derived domains such as that being sketched. This would provide the strongest form of support for the theory. Unfortunately, the interesting implications of this theory, especially as concern the repertoire of possible prosodic/phonological categories, cannot be fully pursued here. Rather the focus will be on the claim that the end-based theory provides a general and unified theory of possible mappings between syntactic structure and prosodic/phonological structure in language.

I should mention before proceeding, however, how one might conceivably derive this sort of prosodic structure not directly with respect to the syntax, but from a representation that contained silent demibeats in various numbers between words which had themselves been inserted with respect to syntactic structure. (Recall the discussion in §1.) (22) shows the structure (21) with silent demibeats added:

(22)



Clearly, prosodic structure could be constructed 'from' such a silent demibeat structure by, say, making the ends of a particular prosodic constituent coincide with two successive sequences of silent demibeats with matching cardinality of a particular magnitude. (Presumably the cardinalities involved would be somewhat restricted, perhaps to only 1, 2 and 3+.) In this way, a hierarchical representation of the sort defined

above is automatically derived. It is an interesting point that the rules for silent demibeat insertion that were proposed in Selkirk (1984a) share the same fundamental property as the prosodic constituent-deriving rules to be proposed here: they appeal only to one or the other end of syntactic constituents of designated types. Both are thus end-based theories of the syntax-phonology mapping. Either the *sdb*-based theory of prosodic structure or the theory that takes prosodic structure as basic will generate the same class of derived phonological constituent structure domains. As I said above, I am opting for the latter theory, with the hunch in mind that in general, as in Chi Mwi:ni, the metrical grid will be constructed with respect to prosodic structure and not with respect to the syntax.

An end-based theory of the sort described here will define a class of possible end-settings ${}_a[$ and ${}_a]$ for derived domains. So far, it's been suggested that the settings in (23) may play a role. There are likely to be a few more (cf. below):

(23) End parameter settings

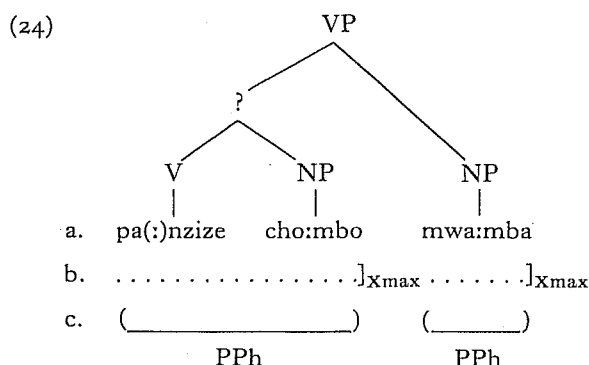
- (i) a. ${}_a\text{Word}$ b. $\text{Word}{}_a$
(ii) a. ${}_a\text{X}_{\text{max}}$ b. $\text{X}_{\text{max}}{}_a$

A setting for the category α and for right or left end (the direction of the bracket) constitute the setting of a parameter in the syntax-phonology domain-deriving component of a language. The choice of a particular parameter setting for α in the grammar of a language, taken in combination with the particulars of phrase structure of that language, defines the syntax of α -derived domains, one class of prosodic constituents, in that language. One imagines that there might be a relation between the gross phrase structure characteristics of a language and the parameters of the syntax-phonology mapping. Is it mere coincidence, for example, that postpositional Japanese includes a word and all following function words in a (minimal) 'accentual phrase' (cf. McCawley 1968; Miyara 1980; Poser 1984), indicating an end-setting of $\text{Word}{}_a$, while English, with the opposite phrase structure characteristics, has the opposite end-setting at the word level? Probably not. Concerning prosodic constituents at a higher level, Nespor & Vogel (1986) suggest that languages will include in the same domain a phrasal head and a complement to the head found 'on the recursive side'. It should be mentioned, however, that two languages whose phonological phrasing is discussed in this paper, Xiamen and Ewe, display at least superficial similarities in their phrase structure properties, though differences in their choice of phrasal end-setting. The intriguing question of what relation there might indeed be between the choice of end-settings for derived domains and the general properties of syntactic structure in the language can unfortunately not be addressed here, though it calls out for investigation.

Chen's work on Xiamen tone sandhi gives an exemplary demonstration of how the setting of the end parameter ${}_a\text{X}_{\text{max}}$, applied to a wide array of surface syntactic structures in the language, gives just the right character-

isation of the relevant derived domain.¹⁰ The reader is referred to this work. Here I want to illustrate the theory with data available from Chi Mwi:ni. Though the data is far less complete than that from Xiamen, it allows us to illustrate the theory with some now-familiar examples. Consider the examples discussed above, which showed that the domain for Chi Mwi:ni stress was not a phrasal constituent of the syntax. If we assume simply that the relevant parameter setting in the mapping is]_{x_{max}} as in Xiamen, we achieve just the right characterisation of the derived domains for stress.

Consider first (24), which corresponds to (16a):

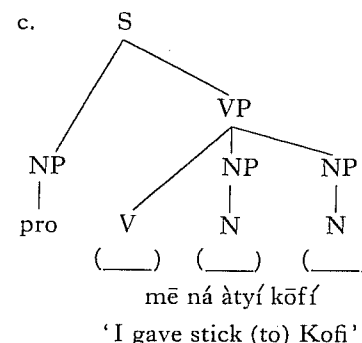
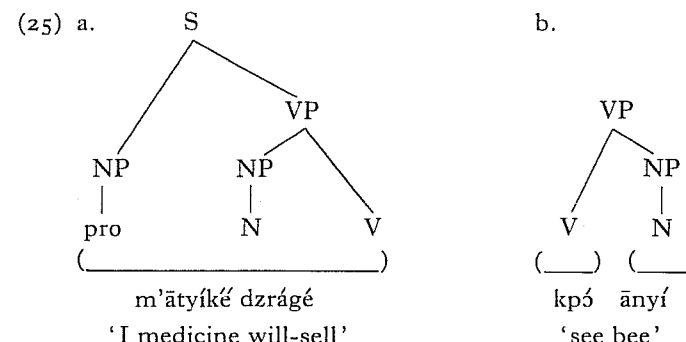


In the syntactic structure (24), the end-setting identifies two right ends of maximal projections, as shown in line b. (Remember that line b is not itself part of any representation.) And the domains in line c, which do form part of the representation, are derived on the basis of this. The domains are named PPh (phonological phrase), following Hayes, though the theory of how these domains are derived here differs from that assumed by Hayes, as we will see below.

(24) illustrates the fact that in a language with the setting $]_{X_{\max}}$ a complement (argument) immediately following a lexical head will be included in a derived domain with that head, whereas a second complement (argument) will never be in the same derived domain with the preceding one. This state of affairs is an extremely familiar one from all studies on the syntax-phonology mapping in languages with head-complement-complement structure (e.g. Selkirk 1972, 1978, 1984a; Rotenberg 1978; Nespor & Vogel 1982, 1986; Kaisse 1985; Hayes 1984; Chen 1985). Note that the same $]_{X_{\max}}$ parameter setting in a language like Xiamen, which allows complements to both precede and follow the verb, puts the verb in a separate domain from its immediately preceding complement but in the same one with a following complement, as shown in (20). This is exactly the right result.

A setting of the $X^{\max}$ parameter in the opposite direction, $x_{\max}[-]$, gives just the domains described by Clements for Ewe tone sandhi. (This analysis is in the spirit of Clements' own end-based analysis.) Like

Xiamen, Ewe may have VPs with a single complement to the left of the verb along with VPs where the complements are located on the right. But in Ewe, given the left- $X^{\max}$ end-setting, it is the complement that precedes the verb which finds itself in the same derived domain with the verb, and any following complements are in separate derived domains. These domains are illustrated in (25):



The sandhi rule at issue is one which makes a mid tone extra high when it is located between two high tones all within the same derived domain. Clements formulates the rule as in (26a); in (26b) I have added the end setting for Ewe that gives the domains within which the play of the rule is confined:

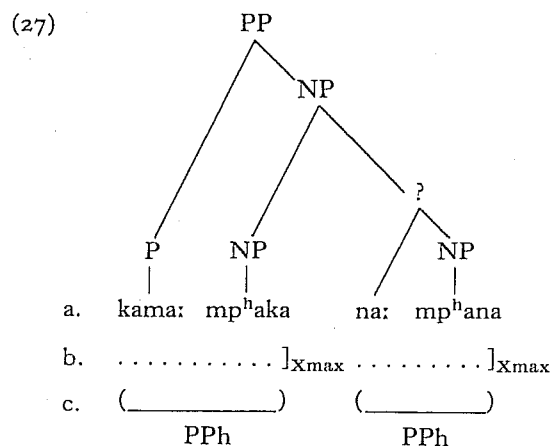
(26) a. *Ewe Tone Sandhi* (Clements 1978)

Mid→Extra High/High—High Tones: – mid
 ' high
 \ low
Fine End Parameter Setting " extra high

b. *Ewe End Parameter Setting*

In (25) we have the (intermediate level) representations of tone that result from the application of this sandhi rule. In (a) we have an extra high between two highs, showing that the rule has applied. In (b) and (c), we see a mid tone located between two highs, the failure of the rule to apply indicating that the tones (and their accompanying words) are not all held within the same derived domain.

It is a highly interesting fact that with the theory of the syntax-phonology mapping under consideration the proper deriving of domains in Chi Mwi:ni, Xiamen and Ewe falls out simply from an end-setting for $X^{\max}$, in one or the other direction. No mention is made in the mapping here of the relations that may obtain between elements belonging to the same domain, be they relations involving the notions head/complement, or c-command, linear precedence or adjacency. Which is to say that the end-based theory is more restrictive than any of the others that will be discussed below, and so, if empirically supported, to be preferred.



Returning to our Chi Mwi:ni examples, (27) illustrates the fact that end-based domains may correspond to a sequence that is not a constituent of syntactic structure. With the setting $]_{X^{\max}}$, an $X^{\max}$ -derived domain simply extends from one instance of $]_{X^{\max}}$ to another (or to the end of the utterance, if there is no further $]_{X^{\max}}$). In (27) that sequence includes the first $N^{\max}$ of a coordinate structure as well as the preceding P. (Note that the same sort of mapping into $X^{\max}$ -derived domains would be obtained if P in (27) were a V or any other lexical category.) It should be pointed out that the non-isomorphism exemplified here is of a limited sort. For example, the preposition *does* c-command the neighbour with which it is joined in the same prosodic domain. In principle, however, an end-based theory could allow for a more radical lack of isomorphism between syntactic and prosodic structure. For example, if a language whose function words were normally to the right of a phrasal head, as in $[fw_x X]_{XP} - [fw_y Y]_{YP}$, were to have a left-end-setting for *word*, $word[$, then

the prosodic word in that language would group the function words from a following phrase with the 'real' word at the end of a preceding phrase, as in $(fw_x)_{wd} (Xfw_y)_{wd}$. It is quite possibly cases of just this sort which are reported in Klavans (1985). Probably end-settings which would give rise to such a radical non-isomorphism would be the marked choice for a language. Nonetheless, the existence of such cases would provide just the sort of evidence which supports the theory that there is a representation of prosodic domains in the grammar, derived via an end-based mapping of the sort being proposed here.

A few further examples from Chi Mwi:ni are given below. (All are drawn from Kisseberth & Abasheikh.) Line a gives the surface pronunciation. As before, the presence or absence of length gives the indication of where the domain ends lie. Line b gives the input to the sentence phonology, which includes the output of the word phonology with all the sources of length represented, and the labelled bracketing of the surface syntax. Line d gives the representation of stress domains for the expression, derived through the application of the $]_{X^{\max}}$ setting to the surface bracketing, as indicated in line c. The stress and shortening rules, applied within the domains so derived, give just the pattern of length we see in line a. The examples in (28), containing the same lexical items, form a minimal pair. The vowel length difference is attributable to a difference in surface syntax which is relevant to the formation of derived domains:

(28)

- i. a. nt^hi: ŋk^havu: 'dry land' ii. a. nt^hi: ŋk^havu: 'the land is dry'
- b. $NP[N[nt^h i:] A[ŋk^h avu:]]_{NP}$ b. $S[NP[nt^h i:]_{NP} AP[ŋk^h avu:]_{AP}]_S$
- c.]X_{max} c.]X_{max}.....]X_{max}
- d. () () ()

In (29) we see that in noun phrases, which are always right-branching, with the head on the left, the head is joined in a stress domain with what follows:

- (29) a. mapendo ya maski:ni 'the love of a poor man'
- b. $NP[mape:ndo: [ya: NP[maski:ni:]_{NP}]]_{NP}$
- c.]X_{max}
- d. ()

In (30) we see that a verb and a following (right-branching) complement NP will together form a domain:

- (30) a. shikani malimu wa:saba 'seize (pl.) the seventh teacher'
- b. $VP[V[shika:ni:]_V NP[ma:limu: [wa: NP[saba:]_{NP}]]_{NP}]_{VP}$
- c.]X_{max}
- d. ()

(31) simply presents another example of the fact that the first complement to a verb will always be set off from a second complement:

- (31) a. nimwandikilile Nu:ru xati 'I wrote Nuru a letter'
 b. $VP[V[nimwandikilile:]_V NP[Nu:ru:]_{NP} NP[xati:]_{NP}]_{VP}$
 c.] X_{max} ] X_{max}
 d. () ()

These examples from Chi Mwi:ni, then, provide further support for the end-based approach to the characterisation of derived domains such as is proposed by Chen for Xiamen and Clements for Ewe. In Chi Mwi:ni and Xiamen a right-end X^{max} parameter setting constitutes a correct theory of the syntax-phonology mapping, in one case giving the domains for (end-sensitive) stress, and the other the domains for tone sandhi. The subtly different phrasing effects in the two languages derive from their phrase structure differences. In Ewe a left-end X^{max} setting, in combination with the particulars of Ewe syntax, gives just the right array of domains for tone sandhi. So we have evidence here which supports the claim that an end-based theory of domains is indeed a (correct) general theory of possible relations between syntactic and phonological structure.

Hayes uses the term PHONOLOGICAL PHRASE to designate the constituent of phonological representation which provides the domain for stress in Chi Mwi:ni. The same term has also been used for smaller sorts of domains, which wouldn't be generable throughout an X^{max} setting. In order that the use of terms for prosodic structure constituents more clearly indicate just what sort of correspondence to the syntax is involved, let us call the constituents derived through an X^{max} setting MAXIMAL PHONOLOGICAL PHRASES, using 'maximal' as mnemonic for X^{max} . By using the term 'maximal', I do not intend to imply (at this point at least) that there could not be larger phonological phrases defined through an end-based mapping, as might be found if there were settings s_l or s_s in the repertoire of parameters. (If there were such, the terminology could be changed.) Yet the lack of evidence in favour of this latter sort of setting suggests that what are being called maximal phonological phrases here may indeed turn out to be the maximal ones (literally speaking) that the theory should allow. Recall that one of the central claims being made with the end-based theory presented here is that the repertoire of possible prosodic/phonological constituents (above the foot and below the intonational phrase) is a function of the different possible end-settings that the theory of grammar makes available in the mapping. It is not likely that there are many: X^{word} settings give us phonological/prosodic words; X^{max} settings give us (maximal) phonological phrases. A smaller type of phonological phrase has been discerned in a variety of languages. It is one which includes the head of a phrase and preceding or following specifier material, but not any complements (arguments) to the head. (Whether the adjunct material

joined with the head either follows it or precedes it, or both, may depend on the language, and also on what is meant by 'specifier adjunct phrase' as opposed to 'non-specifier adjunct phrase'.) Let us call this sort of constituent a SMALL PHONOLOGICAL PHRASE. Below I will argue that there is an end-setting definable in grammar which sets off the head-specifier or specifier-head sequences that correspond to small phonological phrases.

Let us examine a familiar case of small phonological phrases – the domain for 'unmarked' liaison in colloquial French (Selkirk 1972; Nespor & Vogel 1986¹¹). LIAISON is the failure of a final consonant to delete when followed by a vowel at the beginning of the next word. In colloquial French one finds liaison between words located in sequences which include the head of a phrase, preceding modifiers and determiners, and other c-commanding function words, as in (32)–(34):

- (32) (Sais-tu) (quand ils inviteront) (un autre grand artiste)?
 'Do you know when they will invite another great artist?'
 (33) (Ces très aimables enfants) (en ont avalé).
 'These very nice children some-have-drunk.'
 (34) (On m'a souvent amené) (dans un énorme wagon).
 'One often took me in an enormous train car.'

(The domains of liaison are parenthesised. Each instance of liaison is signalled by the underlining of the undeleted consonant.) What I want to argue is that there is an end-setting in the domain-deriving module which gives rise to just these small phonological phrases.

It is known that phrasal constituents in specifier position display a more restricted internal syntax than phrases not in that position. Specifier adjective phrases, for example, may not contain complements to the head. (This generalisation that specifier phrases are not fully-fledged phrases has been made by a number of authors, e.g. Emonds 1985; Stowell 1981.) One way of capturing this fact would be to say that adjectives in the specifier position do not themselves project further phrase levels in the X-bar hierarchy.¹² Which is to say that they are not 'real', i.e. phrasal, heads. And that the phrases that contain them (along with possible modifying adverbials) are not maximal projections. (On this story, a modifier + adjective unit in specifier position, such as *très aimable*, would be considered to be some sort of compound word, not a phrase.) If this characterisation of the syntax of specifiers is correct, then it is a simple matter to define an end-setting in the syntax-phonology mapping which will give rise to small phonological phrases. All we need to do is define two types of words – words that are heads of maximal projections, and words that are not. The first can be designated X^{head} . With this distinction made, and an end setting $]X^{head}$ assumed to form part of the grammar of French,¹³ the appropriate domains for liaison can be straightforwardly derived:

- (35) a. $s[[\text{Sais-tu}]_V [[\text{quand}]_{\text{Comp}} [\text{ils} [[\text{inviteront}]_V [\text{un} [\text{autre}]_{\text{AP}} [\text{grand}]_{\text{AP}} [\text{artiste}]_{\text{N}}]_{\text{NP}}]_{\text{VP}}]]]$
 b. $]_{\text{Xhead}}$ $]_{\text{Xhead}}$ $]_{\text{Xhead}}$
 c. () () ()
- (36) a. $s[[\text{Ces} [[\text{très}]_{\text{Det}} [\text{aimables}]_{\text{A}}]_{\text{AP}} [\text{enfants}]_{\text{N}}]_{\text{NP}} [[\text{en} [\text{ont}]_{\text{Aux}} [\text{avalé}]_V]_{\text{VP}}]]]$
 b. $]_{\text{Xhead}}$ $]_{\text{Xhead}}$
 c. () ()
- (37) a. $s[[\text{On}_{\text{VP}} [[\text{m}' [\text{a}]]_{\text{Aux}} [\text{souvent}]_{\text{AdvP}} [\text{amené}]_V \text{PP} [[\text{dans}]_P [\text{un} [\text{énorme}]_{\text{AP}} [\text{wagon}]_{\text{N}}]_{\text{NP}}]]]]]$
 b. $]_{\text{Xhead}}$ $]_{\text{Xhead}}$
 c. () ()

As the b lines indicate, the $]_{\text{Xhead}}$ setting distinguishes between the heads of maximal projections and the heads of specifier constituents, and so, for example, a specifier adjective and a following head noun are grouped together in a (small) phonological phrase. Moreover, this setting is successful in bringing preceding function words into the small phonological phrase with the following phrasal head. This is no surprise: any theory must assume that function words are not identifiable as constituents of any sort in the syntax. Whether heads or not, the bracketing of function words is 'invisible' to the rules deriving phonological domains. In sum, with independently motivated assumptions concerning the status of function words and the status of specifier adjunct constituents in the grammar, we obtain the correct demarcation of the small phonological phrase in French merely through the application of the parameter $]_{\text{Xhead}}$ in the syntax-phonology mapping.

Obviously this account of the syntax of specifiers requires further elaboration: the implications for syntax and semantics are not insignificant. Unfortunately I am not able to pursue here the non-phonological implications of treating specifier adjunct constituents in this way. There is some additional favourable evidence from phonology which I would like to mention briefly, however. It seems that the assumption that specifier adjunct constituents are not maximal projections has just the right result when it comes to defining the place of these constituents in the domains of tone sandhi in Ewe and Xiamen, which involve the application of an X^{max} end-setting. In Ewe, Clements observes, an adjunct phrase following phrasal head and modifying it is included in the same tonal sandhi domain as the head, as shown in (38):

- (38) a. $s[\text{mé}_{\text{VP}} [\text{V} [\text{kpɔ́-é}]_{\text{AdvP}} [\text{nyúíé}]]]$ 'I see-it well'
 b. $\text{X}^{\text{max}}[\dots]$
 c. ()

(Contrast this fact with the fact that an NP (argument) complement following the verb is in a separate tonal domain; cf. (25).) Clements suggests that such adjunct constituents are not 'real' phrases in Ewe, and

therefore behave this way. Following up on this suggestion in the current theory, by not treating these adjunct constituents as X^{max} we obtain just the right effects: the left X^{max} end-setting of Ewe grammar will not pick out the AdvP and so will derive a domain including both of them, as shown in lines b and c. In Xiamen, adjunct constituents have a similarly special status within domains of tone sandhi, which Chen documents rather thoroughly. I believe that it is also possible to bring the Xiamen data under the sway of the sort of generalisation that I am proposing here. The reader is referred to Chen's discussion.¹⁴ It is not without interest that the same assumption that specifier adjunct constituents are not maximal projections does the right kind of work with respect both to the rules with end-settings that mention X^{max} and to those with the lower setting X^{head} .

3.2 Relation-based theories of derived domains

Previous treatments of derived domains in the prosodic structure framework (Selkirk 1978, 1980; Nespor & Vogel 1982, 1986; Hayes 1984; and others) have not made appeal to the *ends* of designated types of syntactic constituent in the characterisation of the mapping between syntactic structure and particular levels of constituent in the prosodic hierarchy. Rather, they have made crucial appeal to various sorts of *relations* obtaining between the syntactic constituents contained in the same derived phonological domain. Consider for example Hayes' characterisation of the phonological phrase in Chi Mwi:ni:¹⁵

(39) Chi Mwi:ni P-Phrase Formation

- a. In $[\text{X}^0 \text{Y}'' \dots]_{\text{X}''}$, where X^0 is the head of X'' and Y'' is an adjacent complement, the sequence $\text{X}^0 \text{Y}''$ forms a P-phrase.
 b. All clitic groups unaffected by (a) form P-phrases.

Let us first discuss part (a) of this definition of phonological phrase. It invokes the notions COMPLEMENT-TO-HEAD and ADJACENCY (of complement to head). An extension of this approach to the cases of Ewe and Xiamen would require invoking the LINEAR ORDER of the constituents involved. Drawing on these various relational notions, the theory is unnecessarily rich, and so lacks predictive power. For example, a relational theory of this sort would in principle allow for the possibility that head and (non-adjunct) complement form phonological phrases regardless of their linear order. It could also require that two complements of the same head be joined in a same phonological phrase. Cases of these sorts are so far unattested. Should this absence prove systematic, support is proved for the end-based theory, which rules them out. An end-based theory is extremely impoverished, by contrast, and so highly constrained. It allows only for domain-formation rules with parameter settings drawn from the repertoire $_{\alpha}$ [or $_{\alpha'}$. And, as was shown above, a simple end-setting of just this sort – $]_{\text{X}^{\text{max}}}$ – is enough to guarantee just the results for Chi Mwi:ni and Xiamen (and for Ewe with the direction reversed) that complement-to-head, adjacency and linear order combine to produce in the alternative theory.

Given the present state of our knowledge of these matters, the end-based theory, which makes no appeal at all to relations between constituents in sequence in surface syntactic structure¹⁶ is to be preferred as the one with the greater explanatory power.

It's a significant fact that provision (a) of (39) does not alone suffice to characterise the relevant phonological domain in Chi Mwi:ni. Provision (b) is intended to guarantee, among other things, that (i) any head on its own without complements will be treated as a phonological phrase, and (ii) that in a sequence $\text{Prep}_{\text{NP}} [\text{NP conj NP}]_{\text{NP}}$, such as found in (27), the Prep forms part of the PPH with the first conjunct NP. (A CLITIC GROUP for Hayes consists essentially of a content word and any c-commanding function words.) Note that with the end-based theory these fall out without any extra proviso, as was illustrated above.¹⁷ On balance, then, the end-based theory gains in descriptive adequacy over the relation-based account.

In the introductory section of this paper it was pointed out that entertaining a model of sentence phonology including a subcomponent of rules which are *directly* syntax-sensitive and another subcomponent which are only *indirectly* syntax-sensitive (via their sensitivity to syntax-dependent prosodic structure) naturally gives rise to this question for any seemingly syntax-sensitive rule: which of the two subcomponents does it belong to? Some scholars (notably Rotenberg 1978 and Kaisse 1985) have claimed a very large role for the earlier syntax-sensitive subcomponent in sentence phonology, and have implicitly denied any significant role (if any at all) for a syntax-derived prosodic structure and the accompanying subcomponent of P-structure-sensitive rules. Kaisse, for example, has argued that it is a general property of rules of external sandhi (sandhi rules of the sentence phonology) that they are restricted in their application essentially to words in sequence where the relation of C-COMMAND (more precisely, DOMAIN-C-COMMAND: see below) holds. I would like to argue here that the class of rules claimed by Kaisse to be subject to this condition need to be understood instead as P-structure-sensitive rules, ones which apply on prosodic domains derived through an end-based mapping such as that being proposed here. The argument I am making is not that there are no rules of sentence phonology that are directly governed by syntactic structure, rather only that one class of cases claimed to be so is not.

Kaisse defines domain-c-command as follows (Kaisse 1985: 159):

(40) *Domain-c-command*

in the structure $[_{X^{\text{max}}} \dots \alpha \dots]$, X^{max} is defined as the domain of α .
Then α c-commands any β in its domain.

Her proposal is that a language will impose the requirement of domain-c-command on the operation of a sandhi rule of sentence phonology in one of the following two ways:

(41) Kaisse (1985)

For a rule to apply to a sequence ab of two words a and b

- (i) a must domain-c-command b or
- (ii) b must domain-c-command a

In her treatment of Ewe tone sandhi, where tone sandhi affects the words in (25a) but not (25b, c), the second option is claimed to have been taken. This achieves the correct results for the Ewe sandhi. In (25a) the Verb domain-c-commands the preceding Noun and sandhi is allowed, whereas in (25b, c), the interference of the domain-defining X^{max} node NP prevents the N on the right from domain-c-commanding the preceding word, and sandhi is blocked. (It should be obvious why the notion of domain-c-command, with its appeal to maximal projection, is giving results with respect to the domain of tone sandhi quite parallel to the prosodic structure domain theory proposed here with its $]_{X^{\text{max}}}$ end-derivation.)

There are three criticisms of Kaisse's theory that I wish to advance here. The first is that, in a framework like Kaisse's, wherein rules of sentence phonology are directly governed by the syntax, the notion domain-c-command on its own cannot provide a general unified theory of how syntax impinges on phonology. In particular it gives no representation of the *ends* of non-syntactic constituent domains such as are required for the application of end-sensitive rules like the stress rules in Chi Mwi:ni. The domain-c-command condition is no help in directing the Chi Mwi:ni stress rule to the ends indicated in (24). Indeed, the domain-c-command theory was not designed with end-sensitive rules in mind. Kaisse casts it explicitly as a theory of external sandhi rules, involving two words in sequence within the sentence. Yet the derived domains that must be defined for external sandhi rules are no different from those that must be defined for end-sensitive rules, as the discussion of Chi Mwi:ni, Ewe and Xiamen has shown. One wants a theory that can deal with both in a unified way.

To salvage the c-command theory as a general theory of the syntax-phonology interface, one might wish to conceive of it as a theory of the conditions governing the mapping between syntactic structure and prosodic structure. Conceived in this way, the (asymmetric) domain-c-command condition, parameterised as 'in a sequence ab , a domain-c-commands b /b domain-c-commands a ', would be required to hold of any two successive elements contained in a same derived domain D .¹⁸ Phonological rules would not themselves be subject to the domain-c-command restriction, but would only be sensitive to prosodic structure domain, applying across them, or at their ends, in the usual way. For the Ewe case, it is easy to see that the parameter setting ' b domain-c-commands a in ab ' would give just the same definition of derived domains as the X^{max} setting of the end parameter in an end-based theory of derived domains.

But now the domain-c-command theory of the mapping of syntax into derived prosodic structure domains is subject to the same sort of criticism levelled above against the other relation-based theories of the mapping. Notice that the theory invokes three distinct sorts of relation obtaining between the syntactic constituents within the derived domains: (i) C-COMMAND, (ii) ADJACENCY and (iii) LINEAR ORDER. An end-based theory such as the one being advocated here simply identifies *ends* of syntactic constituents and achieves the same result. In defining the same sort of object, the c-command theory and relation-based prosodic structure

theories of the sort discussed above proceed by defining the 'insides' of that object, invoking this variety of relational notions. Only derivatively are the ends defined. The end-based theory simply defines the ends, and gets the 'insides' for free. It is the simpler and more explanatory of the two.

A final criticism of the domain-c-command theory is that it fails to give a place to the hierarchy of (derived) phonological domains that have been shown to be at play in sentence phonology. What is the analogue of the prosodic structure domains PROSODIC WORD or SMALL PHONOLOGICAL PHRASE discussed above? While it may indeed be possible to construe rules applying within such domains as being subject to syntactic conditions, the onus is on the advocate of a 'phonosyntactic' characterisation of these rules to show that a theory as simple and general as the end-based theory of prosodic domains being defended here is available for the entire class of cases.

4 On ends in sentence phonology

In the preceding sections an additional argument has been adduced from Chi Mwi:ni that the phonological representation of the sentence, P-structure, must (in part) be analysed as a sequence of prosodic constituents which may serve as the domains for rules of the sentence phonology. In Chi Mwi:ni the rule at issue is an end-sensitive stress rule. The derived domains whose ends are relevant for stress are themselves claimed to be defined with respect to the ends of constituents of syntactic structure. Clearly the notion END is playing a central role in defining how rules in sentence phonology are sensitive to the higher order structure of the representation at hand, whether in the syntax-phonology mapping, where the relevant representation is syntactic, or in the phonology *per se*, where the constituency is phonological.

Is it possible that ends are the only sort of information about constituency that can be accessed in the sentence phonology? (Note that with the ends of derived domains we also define automatically the SPANS of the string located between two ends of the same sort.) Is it possible that relations between constituents in sequence never play a role in the sentence phonology? For P-structure rules, this would mean that no 'domain-juncture' rules (Selkirk 1980a), which are restricted to configurations $[_\alpha \dots]_\beta [_\gamma \dots]_\gamma$, where α , β and γ are phonological constituents (α and β possibly identical), are available. And indeed, according to Vogel (1984), this seems to be the case.¹⁹ For rules directly sensitive to syntactic structure this would mean that no relational notions like c-command, etc., could have a role. With my argument here concerning the proper characterisation of derived domains, headway has been made towards a principled excluding of such relational notions from the syntax-dependent component of the sentence phonology. At this point, however, we must resist going too far. Until it is known what in general the properties are of all the rules which legitimately *do* have a place in the pre-P-structure subcomponent of the

grammar, no extreme conclusions about the possible effect of syntax can be come to.

What can be said, though, is that the evidence presented in the current paper is consistent with a radical restriction on the availability of higher order structural information to rules in two parts of the sentence phonology – in the syntax/prosodic structure mapping and in the phonology *per se*. This is a not uninteresting result.

NOTES

- [1] In Selkirk (1983, 1984a, 1985) I argued that the input to the sentence phonology was not simply the surface syntactic structure, but rather an 'intonated surface structure', consisting of surface structure plus a representation of intonational phrasing and of the assignment of pitch accents to individual words of surface structure. This intonated surface structure was argued to form the input to logical form, on the one hand, wherein an interpretation would be assigned to the sentence partly by virtue of its intonational structure, and the input to the sentence phonology on the other, where, it was argued, the intonational structure contributed to determining various aspects of the prominence relations (stress pattern) of the sentence. In the current paper, in order to simplify the exposition and to raise the relevant points about the organisation of sentence phonology more clearly, I have ignored the distinction between surface structure and intonated surface structure.
- [2] An alternative suggestion by Kisseberth & Abasheikh is to treat word-final lengthening as a rule of the sentence phonology, restricted to words that are not phrase-final. This would obviate the need for a phrase-final shortening. It would not substantially detract from the strength of the argument to be presented below for a stress-based analysis of phrasal shortening. The gains of such an analysis are dubious. In particular, it would require an (extrinsic) statement of the ordering of the stress rule and the word-final lengthening rule. Moreover, given a stress-based analysis of phrasal shortening, no other simplification in the grammar is accomplished by adding this medial restriction to the rule and putting it in the sentence phonology. The same stress-based analysis will do its work regardless.
- [3] According to Prince (1983) and Selkirk (1983b) the binarity of stress patterns results not from counting to two, but rather from a maximal filling out of a perfect grid alternation, one which avoids both clash and lapse.
- [4] Goodman does use the term 'stress' for what Kisseberth & Abasheikh call 'pitch accent' in Chi Mwi:ni. Pitch accents, or high tone, appear on either the penult or final syllable of a word or phrase, depending on a variety of factors, yet to be fully described.
- [5] I might add that there is evidence as well that this phrasal domain is also a domain within which certain syllabic phonotactic constraints must be allowed to hold. In particular, Chi Mwi:ni does not allow a long vowel to be followed by a consonant cluster, except if the cluster consists of homorganic nasal + consonant. This would seem to indicate that the Chi Mwi:ni rhyme can have at most two positions, and that, except in the nasal cases, a sequence of two consonants divides into separate syllables. There is data from Kisseberth & Abasheikh which suggests that these phonotactics hold on the phrase. When a word of the form CCVCV(C) is phrase-final, it will in effect 'throw' stress onto the preceding word-final syllable, which would remain long, unless the presence of the following CC sequence induced a syllabification of the following C into the word-final syllable, which would result in the shortening of that vowel. This is in fact what one observes in the cases of this sort that appear in the data, e.g. *chint^hu shpiya* 'something new', not *chint^hu : shpiya*. So it seems that syllabification and stress go hand in hand in applying on this larger-than-word domain.
- [6] An alternative approach, cast in the framework of Lexical Phonology and Mor-

phology, would be to specify that stress rules are crucially restricted to some level α in the 'lexicon'. With this approach, bracketing needn't form part of the structural description of the rules, nor need it form part of the representation being applied to. The domain ends which the stress rules need as reference points are simply the ends of that part of the phonological representation currently under consideration in level α . While viable (within the framework of Lexical Phonology and Morphology) as long as stress is restricted to word-internal domains, this 'bracketless' approach to domain cannot be extended to cases where the domain is not 'lexical', but rather 'phrasal' or 'sentential'. This would be true if the domain of stress α were a constituent of syntactic structure, since syntactic structure cannot be analysed in terms of levels in the sense of Allen (1978), Kiparsky (1982, 1985) and Siegel (1974). (Indeed it is precisely this distinction between syntax and morphology that Lexical Phonology and Morphology seeks to exploit.) And it would be all the more true if the phrasal domain α were not a constituent of the syntax, but rather of the phonology. Here there would be no getting round the *representation* of domain in the phonological representation of the sentence (via some sort of bracketing) and an appeal to α in the *structural description* of the rules.

- [7] See Nespor & Vogel (1986) for examples where clitic groups are the domain of assignment of stress. The examples they offer involve only final or penult (possibly final, with extrametricality) stress in clitic groups.
- [8] There *are* cases where a cliticisation of function words must be assumed. In Selkirk (1972, 1984a) I argue that in English, pronouns must cliticise to a preceding Verb or Prep. Without this assumption their phonology cannot be derived, while with it, their phonology falls out with no extra stipulations.
- [9] According to Hayes (1984) and Nespor & Vogel (1986), a clitic group contains a content word and adjacent function word which bear the relevant syntactic relation to the content word. That relation amounts to c-command. In the present paper no distinction between clitic group and prosodic/phonological word is being drawn. Any properties strictly attributed to words without clitics are claimed not to involve prosodic constituents in the sentence phonology, but rather to be a matter of the status of the constituent word in the word phonology.
- [10] This is an oversimplification of Chen's position. See note 14, where Chen's conception of the role of specifier adjunct phrases in Xiamen sandhi domains is discussed.
- [11] Nespor & Vogel call this domain a phonological phrase. In Selkirk (1972) this domain corresponded to one where only a single word boundary, #, intervened between the words in sequence. There have been a number of treatments of French liaison in a framework where no such intermediary phonological representation plays a role, e.g. Rotenberg (1978); Kaisse (1985); Lobeck & Kaisse (1984). Comments on this general approach are given in §3.2 below.
- [12] It is crucial here to say what constitutes 'specifier position'. Possibly sisterhood of the adjacent constituent to the head is required. This would distinguish pre- and post-nominal adjective constituents in French, the latter being sister to N-bar, not the head N itself.
- [13] Probably the choice of direction for the X^{head} end setting will always have to give groupings of head with the specifier, and so will depend on the direction in which the specifiers are located with respect to the head in the overall phrase structure of the language. Dependencies of this sort between phonological phrasing and the general phrase structure characteristics of a language are explored by Nespor & Vogel (1986).
- [14] The generalisation that Chen arrives at is that adjunct constituents which are (i) contained within a maximal projection X^{max} and (ii) c-command its head X^1 do not themselves qualify as maximal projections with respect to the end setting $]_{X^{\text{max}}}$ which seeks out the appropriate ends for derived tone sandhi domains. Condition (i) simply distinguishes specifier adjunct constituents from others, as we would expect. The second condition covers two facts. The first is that *conjoined* specifier adjuncts *are* treated as maximal projections by the domain-deriving rule. Could

it be that it is a property of phrasal conjunctions that the 'maximal' analysis is imposed, 'top down', on the component conjuncts? The second fact is that adjunct phrases which have an attached particle *-e-*, though they be in specifier position, are also treated as maximal projections. Possibly the particle itself projects a maximal phrase (taking the adjunct constituent as its complement), and so is itself responsible for inducing the phonological phrasing observed. Or possibly, the property of c-command is necessary to designating an adjunct constituent as a specifier in the first place, and so necessary to ascribing to it the special properties the constituent has as a unit in effect 'outside' the X-bar phrasal hierarchy.

- [15] Similar definitions of comparable domains are given in Selkirk (1978) and Nespor & Vogel (1982, 1986). Hayes' formulation is selected as illustrative of the general approach because it addresses the phrasing issue in Chi Mwi:ni, our chosen language of illustration in this article.
- [16] It is true that the notion HEAD plays a role in the end-based theory. But notice that this is a notion concerning only the relation of a constituent to what dominates it, and is to be distinguished from the syntactic relations in question here, which share the property that they are defined between constituents *in sequence*, not dominating each other.
The other relational notion that played a role in our discussion above was ADJUNCT (TO-HEAD). But recall that this notion was not seen to play a direct role in the end-based mapping itself. Rather, the claim is that adjuncts that are c-commanded by the heads they modify (i.e. specifiers) are not in fact maximal projections. This is properly speaking a claim about the syntax of these expressions. If this claim holds up in the syntax, it permits us to get away without appeal to the notion ADJUNCT in the syntax-phonology mapping. Instead it is merely the lack of status as a maximal projection that explains the special properties of specifier adjuncts for the sentence phonology.
- [17] A further elaboration of this relation-based theory would be required to account for the case parallel to this last one where instead of a function word like a Prep, it were a main verb preceding the conjoined NP. On the face of it, the prediction of the Hayes theory is that the verb with its entire following (conjoined) NP complement (consisting of two NPs) be included in the same PPh. The prediction of the end-based theory is, as sketched above, that the conjunct divide (minimally) into two phrases, and that the verb will join with (material from) the left-hand conjunct to form a phonological phrase. The data available unfortunately do not allow us to check out the prediction.
- [18] To account for the special behaviour of adjunct phrases, an assumption similar to ours would have to be made, according to which they are not maximal projections when specifiers. This move would not alone guarantee the right results, however. The definition of c-command would also have to be modified such that non-maximal projection nodes would not 'count' in determining whether the c-command relation held between two words. For the c-command proposal to be viable in the first place, this revised notion of c-command must be viable in syntax. I have not attempted to ascertain whether it does indeed hold up or not.
- [19] Vogel points out that in the cases where something like domain-juncture rules appears to be at play all involve the juncture of *words*. I presume that the associated phonological effects derive from the status of words in the word phonology, and not from their juncture in the sentence phonology.

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